Tucker order-3 vs order-4

node207-23, H100 80GB, BF16 autocast, Llama geometry, seq 1024

16,384 tokens per optimizer step

1 Parameter count

Budget	Layout	Parameters	vs dense
133M	Tucker-4	132,998,816	-124,190,048
133M	Tucker-3: split input	133,002,112	-124,186,752
133M	Tucker-3: split output	132,999,936	-124,188,928
133M	Tucker-3: paired balanced	132,999,680	-124,189,184
225M	Tucker-4	225,023,872	-32,164,992
225M	Tucker-3: split input	225,000,704	-32,188,160
225M	Tucker-3: split output	224,997,120	-32,191,744
225M	Tucker-3: paired balanced	224,999,552	-32,189,312

2 Forward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	609.6	50.0
133M	Tucker-3: split input	633.7	103.5
133M	Tucker-3: split output	629.5	92.4
133M	Tucker-3: paired balanced	623.9	55.5
225M	Tucker-4	600.7	58.4
225M	Tucker-3: split input	616.6	111.9
225M	Tucker-3: split output	626.4	102.2
225M	Tucker-3: paired balanced	618.8	56.4

3 Backward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	777.3	88.6
133M	Tucker-3: split input	743.6	149.6
133M	Tucker-3: split output	760.1	173.4
133M	Tucker-3: paired balanced	530.3	74.5
225M	Tucker-4	760.0	114.5
225M	Tucker-3: split input	724.8	175.3
225M	Tucker-3: split output	749.6	206.0
225M	Tucker-3: paired balanced	522.0	77.8

4 Optimizer, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	88.1	68.0
133M	Tucker-3: split input	86.4	70.8
133M	Tucker-3: split output	85.0	60.1
133M	Tucker-3: paired balanced	81.9	57.8
225M	Tucker-4	90.8	71.4
225M	Tucker-3: split input	84.0	72.1
225M	Tucker-3: split output	96.8	86.5
225M	Tucker-3: paired balanced	81.6	69.3

5 Retraction, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	17.5	17.9
133M	Tucker-3: split input	95.8	96.5
133M	Tucker-3: split output	110.5	109.6
133M	Tucker-3: paired balanced	28.4	28.5
225M	Tucker-4	23.9	23.8
225M	Tucker-3: split input	384.1	384.1
225M	Tucker-3: split output	363.8	363.3
225M	Tucker-3: paired balanced	52.3	53.1

6 Full step, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	1491.6	225.4
133M	Tucker-3: split input	1561.7	421.3
133M	Tucker-3: split output	1589.5	436.5
133M	Tucker-3: paired balanced	1265.2	217.2
225M	Tucker-4	1476.3	269.6
225M	Tucker-3: split input	1812.8	744.6
225M	Tucker-3: split output	1844.9	759.7
225M	Tucker-3: paired balanced	1278.7	258.0

7 Forward+backward peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
133M	Tucker-3: paired balanced	3.75	9.63
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	4.19	9.72
225M	Tucker-3: split output	4.20	9.70
225M	Tucker-3: paired balanced	5.34	10.84

8 Full-step peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
133M	Tucker-3: paired balanced	3.75	9.63
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	5.60	9.72
225M	Tucker-3: split output	4.51	9.70
225M	Tucker-3: paired balanced	5.34	10.84

9 Fastest order-3 full step

Budget	MBxAcc	Layout	ms	time/Tucker-4
133M	1x16	Tucker-3: paired balanced	1265.2	0.848x
133M	16x1	Tucker-3: paired balanced	217.2	0.964x
225M	1x16	Tucker-3: paired balanced	1278.7	0.866x
225M	16x1	Tucker-3: paired balanced	258.0	0.957x

10 Run

Field	Value
Commit	`2c55c11`
Run	`node207-23/gpu3`
GPU	NVIDIA H100 80GB HBM3
PyTorch	2.9.1+cu128
CUDA	12.8
Samples	3 warmup + 12 measured

Every order-3 layout has three trainable factors and one fixed singleton buffer. Paired balanced groups three exact input/output factor pairs, pre-packs a BF16 operator, and reuses saved expansion intermediates in backward. Split-input/output leave one side unsplit. Ratios above 1 mean slower than Tucker-4.